

The Rotating Solution as the Quarter-Turn Dual of the Static Field

gravitomagnetism, the horizonless rotating object, and the slip-core hand-over (ledger D7)

Evaluation note — FRC 21-gravity

June 2026

Abstract

D7 has two parts: the rotating strong-field solution and the inward slip-core hand-over. Both are the quarter-turn. A rotating mass carries its angular momentum on the boost cycle of the frame; its momentum flux is the off-diagonal of the dust tensor, the quarter-turn dual of the rest mass, and it sources a *gravitomagnetic* potential that is the quarter-turn dual of the static gravitoelectric field — the same Q_4 that supplies the conjugate momentum of the radiative sector (D5) and the conjugate-chart registration of the deep regime (D4). The frame-dragging recovers Lense–Thirring and extends to the strong field through the exponential reading; the rotating object is horizonless, with only the redshift floor, so it carries an ergoregion at higher spin but no event horizon. The shadow splits prograde from retrograde at order a , leaving low-spin Sgr A* the clean target, and the rotating background now exists for a quasinormal ringdown. The slip-core hand-over is the rotation into the conjugate (spectral) chart. The residue is the exact nonlinear (rapid-rotation) metric.

1 Gravitomagnetism is the quarter-turn dual

The static field is the gravitoelectric potential $u = Gm/c^2r$. A rotating cluster carries angular momentum J : its state $w = a + b\eta$ on the boost cycle has a non-zero momentum component, and the dust tensor’s off-diagonal m_{ab} (the body’s **Prop:source**) is the momentum flux. That flux sources a gravitomagnetic vector potential exactly as a current sources the magnetic field; the two are dual under the quarter-turn Q_4 , the finite Fourier rotation that exchanges the gravitoelectric and gravitomagnetic readings of one field.

Proposition 1 (Frame-dragging). *The gravitomagnetic dipole $A_\varphi = 2GJ \sin^2 \theta / (c^2 r)$ gives the frame-dragging angular velocity*

$$\omega(r) = \frac{2GJ}{c^2 r^3},$$

*the Lense–Thirring rate of the body’s Proposition **prop:dragging**. In the strong field the gravitoelectric part carries the exponential reading $e^{\mp u}$ and the gravitomagnetic part composes with it, giving the slow-rotation metric*

$$ds^2 = -e^{-2u} c^2 dt^2 + e^{2u} \delta_{ij} dx^i dx^j - 2\omega r^2 \sin^2 \theta e^{2u} d\varphi c dt.$$

2 Horizonless, with an ergoregion

Remark 1 (No event horizon). As in the static case, $g_{rr} = e^{2u}$ is finite for all $r > r_*$ and the only special surface is the operational floor $r_f = r_s / \ln \Omega$, where the redshift reaches $\Omega^{-1/2}$. There is no coordinate singularity and no event horizon. At sufficient spin the frame-dragging makes the static Killing vector spacelike in a thin shell outside r_f — an *ergoregion* without a horizon, which supports superradiant energy extraction and a (model-dependent) ergoregion instability, a signature distinct from Kerr. Its exact location is an order- a^2 datum of the nonlinear solution.

3 Observables

At leading order in $a = J/Mc$ the spin lifts the prograde/retrograde degeneracy of the photon ring: the static value $b_c = 2e Gm/c^2$ (the +4.63% shadow) splits, displacing the shadow centroid at $O(a)$ while the mean diameter moves at $O(a^2)$. The magnitude is the Kerr scale (up to a few percent), so low-spin Sgr A* remains the clean arena. Crucially the rotating background now exists, so the ringdown of a rotating remnant can be computed as a genuine quasinormal-mode problem on this metric, rather than only through the eikonal shadow correspondence — the upgrade a referee asked for.

4 The slip-core hand-over

The slip core $r_* = \sqrt{r_g \ell_P}$ is the scale half-turn (the body’s Remark `rem:halfturn`): the log-midpoint where the frame’s chart rotates fully from position to spectrum. Inside r_* the description is therefore the conjugate (spectral) chart — the very chart in which the registration of the deep regime (D4) and the conjugate momentum of the radiative sector (D5) already live. The hand-over is the quarter-turn, now a derived structure rather than a deferral; the dynamics inside the core is the spectral reading, developed in the framework’s quantum theory.

Status. Ledger D7 moves from O to T|B: the gravitomagnetic (slow-rotation) solution, the horizonless structure, the leading spin observables, and the slip-core hand-over are derived, all on the one quarter-turn duality Q_4 . The residue is the exact nonlinear rapid-rotation metric (the ergosphere location, the high-spin shadow), the Kerr-analogue counterpart of the static nonlinear completion. Checks in `validation/validate_rotating.py`.